

CRD Advanced: Contrasts and Multiple Comparisons

Multiple Comparisons and Post-Hoc Tests

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Learning Objectives

- â€¢ Understand when to use planned contrasts vs. post-hoc comparisons
- â€¢ Apply orthogonal contrasts to test specific hypotheses in CRD
- â€¢ Implement multiple comparison procedures: Tukey HSD, Dunnett's test
- â€¢ Use emmeans package for estimating marginal means and contrasts

Topics Covered

- â€¢ CRD Model Review and ANOVA Assumptions
- â€¢ Diagnostic Plots: Normality, Homogeneity of Variance, Independence
- â€¢ Radon Release Experiment: Real Data Example
- â€¢ Planned Contrasts and Orthogonality Testing
- â€¢ Estimated Marginal Means with emmeans
- â€¢ Tukey's Honest Significant Difference Test
- â€¢ Compact Letter Displays for Multiple Comparisons
- â€¢ Dunnett's Test: Treatment vs. Control
- â€¢ tidyverse Integration with rstatix
- â€¢ Publication-Ready Plots with ggpubr

Key Statistical Concepts

ANOVA Model: $y_{ij} = \mu + \tau_i + \epsilon_{ij}$

Familywise Error Rate: $\alpha_{FW} = 1 - (1 - \alpha)^k$

Effect Sizes: eta-squared, omega-squared, epsilon-squared

For complete interactive content with R code examples, visualizations, and speaker notes, please see the HTML version at:
https://samuelbfernandes.github.io/agst50104-lectures/lecture05_week03.html